Divyesh Sankhla

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SUMMARY

Embedded Software Engineer with several years of experience developing C/C++ firmware and Linux services for production camera platforms. Delivered multithreaded modules, OTA update infrastructure for 1,000+ deployed devices, and 15+ ONVIF/WebSocket APIs integrated with NVR/VMS systems. Skilled in Linux kernel development, device drivers, RTOS, RTSP, WebRTC, and networking protocols for reliable, high-performance embedded products.

SKILLS

Programming: C, C++, Embedded C, Python, Bash, SQLite3, data structures, algorithm analysis, Lean4
Embedded Linux & Kernel: Embedded Linux, Linux kernel modules, character drivers, platform drivers, Device Tree, memory-mapped I/O, DMA, IOVA, IOMMU, cache coherency
RTOS, Architecture & Hardware: Zephyr, FreeRTOS, STM32, ARMv7/v8, UART, RS485, I2C, SPI, ISR handling, IRQ handling, timing analysis, power management, U-Boot, QEMU, systemd, ROS, CUDA
Concurrency & Systems Programming: POSIX threads, IPC, mutex, semaphore, spinlock
Network Protocols & Media: TCP/IP, UDP, POSIX sockets, RTSP, RTP/RTCP, H.264, WebRTC, MQTT
Tools & Workflow: gdb, Valgrind, ftrace, dmesg, tcpdump, Wireshark, netstat, ss, oscilloscope, logic analyzer, Linux serial tools, Git, Make/CMake, Jira, Confluence, Agile/Scrum, code review, Symmetric/Public-Key Cryptography

EXPERIENCE

University at Buffalo — Buffalo, NY
Research Assistant — Oct 2025 - Apr 2026

- Authored Lean4 proofs and Mathlib libraries for asymptotic notation, recurrence relations, Master's Theorem, and algorithm correctness, enhancing reliability and accelerating verification in academic research and teaching.

eInfochips, An Arrow Company — Ahmedabad, India
Embedded Software Engineer — Aug 2021 - Dec 2024

- Built multithreaded embedded C/C++ modules for 5+ Linux IP camera SKUs across streaming, recording, device management, OTA, and systemd runtime while collaborating with 50+ team members.
- Delivered 15+ ONVIF/WebSocket APIs with gSOAP, Lighttpd, POSIX threads, and NVR/VMS integration.
- Diagnosed TCP/IP socket issues in Embedded Linux camera services using netstat, ss, tcpdump, and Wireshark.
- Refactored multi-sensor architecture into a shared codebase across 3 series and SKUs, simplifying maintenance.
- Delivered OTA update infrastructure for 1,000+ deployed devices with staged downloads, A/B partition switching, rollback recovery, checksum validation, image verification, version gating, audit logging, and health monitoring.
- Resolved a time-sensitive UDP RTP/H.264 packetizer defect using Wireshark, restoring low-latency WebRTC streaming over STUN/TURN and reducing decoder frame drops to 0.
- Designed multi-client Live555 RTSP/WebRTC paths with multithreaded C/C++, libdatachannel, Mosquitto signaling, RTP pacing, jitter buffers, and per-client sessions, enabling multiple users on one live stream.
- Engineered Cloud Video Recorder with encrypted RTSPS push, event recording, 30-second fallback buffer for network loss, MQTT configuration for 4+ SKUs, and entitlement control for 5+ features.

Krossmark Innovations — Ahmedabad, India
Embedded Engineering Intern — Jun 2020 - Jul 2020

- Led the serial communication stack for a 5-member sensor-data project, implementing RS485/UART packet framing, CRC validation, retry logic, and link debugging for reliable low-speed links up to 1,200m.

PROJECTS

Linux Kernel and Driver Development

- Developed Linux character and platform drivers with /dev interfaces, ioctl, poll/select, IRQ handling, deferred work, wait queues, Device Tree bindings, and userspace tests, strengthening driver concepts in DMA, IOVA, IOMMU, cache coherency, and kernel module lifecycle.

Bootloader and U-Boot Bring-Up

- Customized U-Boot boot flow in QEMU by adding a board command, tuning bootargs, staging kernel/Device Tree/initrd images, inspecting memory layout, and tracing reset-to-kernel handoff failures.

Zephyr RTOS Firmware and Driver Development

- Implemented multithreaded firmware and low-level drivers with Device Tree integration, priority-based scheduling, ISR-safe interrupt handling, deferred work queues, timing/latency analysis, and basic power-management support.

Pintos Operating System Kernel

- Extended Pintos kernel in C with non-busy-wait alarm clock, priority scheduling, nested priority donation, 4.4BSD-style MLFQS, system calls, ELF loading, file I/O, exec/wait, and safe user memory access.

EDUCATION

University at Buffalo — Buffalo, NY
Master of Science in Computer Science and Engineering, GPA: 3.97/4.00 — Jan 2025 - May 2026

Nirma University — Ahmedabad, India
Bachelor of Technology in Electronics and Communication Engineering — Jul 2017 - Jun 2021